

Import Data

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive")

```
import pandas as pd
df = pd.read_pickle('/content/drive/MyDrive/CPE393/consumer_complaint_dataset.data', compression='gzip')
```

EDA

```
df.head()
```

	topic	input	
0	Debt collection	transworld systems inc. \nis trying to collect...	
1	Credit reporting, credit repair services, or o...	I would like to request the suppression of the...	
2	Debt collection	Over the past 2 weeks, I have been receiving e...	
3	Credit reporting, credit repair services, or o...	I HAD FILED WITH CFPB ON XX/XX/XXXX19 TO HAVE ...	
4	Credit reporting, credit repair services, or o...	I have several accounts that the balance is in...	

```
df.shape
```

```
(492255, 2)
```

Preprocess Text

```
df['sum'] = df['topic'] + ' ' + df['input']
df['sum'].head()
```

	sum
0	Debt collection transworld systems inc. \nis t...
1	Credit reporting, credit repair services, or o...
2	Debt collection Over the past 2 weeks, I have ...
3	Credit reporting, credit repair services, or o...
4	Credit reporting, credit repair services, or o...

```
dtype: object
```

```
input_text = df['sum'].tolist()
```

```
import nltk
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('wordnet')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
True
```

```
import re
from nltk.corpus import stopwords, wordnet
from nltk.stem import WordNetLemmatizer
from nltk import pos_tag

lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))

# remove masked info
def remove_masking(text):
    text = text.lower()
    masked_pattern = re.compile(r'xx+', re.IGNORECASE)
    return masked_pattern.sub('', text)

# POS tag
def get_wordnet_pos(tag):
    if tag.startswith('J'):
        return wordnet.ADJ
    elif tag.startswith('V'):
        return wordnet.VERB
    elif tag.startswith('N'):
        return wordnet.NOUN
    elif tag.startswith('R'):
        return wordnet.ADV
    else:
        return wordnet.NOUN

# clean + tokenize + lemmatize
def clean_text(text: str):
    text = remove_masking(text)
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = text.split()
    pos_tags = pos_tag(tokens)

    lemmatized = [
        lemmatizer.lemmatize(w, get_wordnet_pos(tag))
        for w, tag in pos_tags if w not in stop_words
    ]
    return lemmatized

# ใช้กับ list ของข้อความ
def preprocess_text(text_list):
    return [clean_text(text) for text in text_list]
```

```
text = preprocess_text(input_text)
```

```
with open("/content/drive/MyDrive/CPE393/preprocess_text.txt", "w", encoding="utf-8") as f:
    for doc in text:
        f.write(" ".join(doc) + "\n")
```

✓ Import Preprocess Text From Lab 5

```
file_path = '/content/drive/MyDrive/CPE393/preprocess_text.txt'

with open(file_path, 'r', encoding='utf-8') as f:
    lines = [line.strip() for line in f.readlines()]
df['sum'] = lines
df.head()
```

	topic	input	sum	
0	Debt collection	transworld systems inc. \nis trying to collect...	debt collection transworld system inc try coll...	
1	Credit reporting, credit repair services, or o...	I would like to request the suppression of the...	credit report credit repair service personal c...	
2	Debt collection	Over the past 2 weeks, I have been receiving e...	debt collection past week receive excessive am...	
3	Credit reporting, credit repair services, or o...	I HAD FILED WITH CFPB ON XX/XX/XXXX19 TO HAVE ...	credit report credit repair service personal c...	
4	Credit reporting, credit repair services, or o...	I have several accounts that the balance is in...	credit report credit repair service personal c...	

✓ K-mean

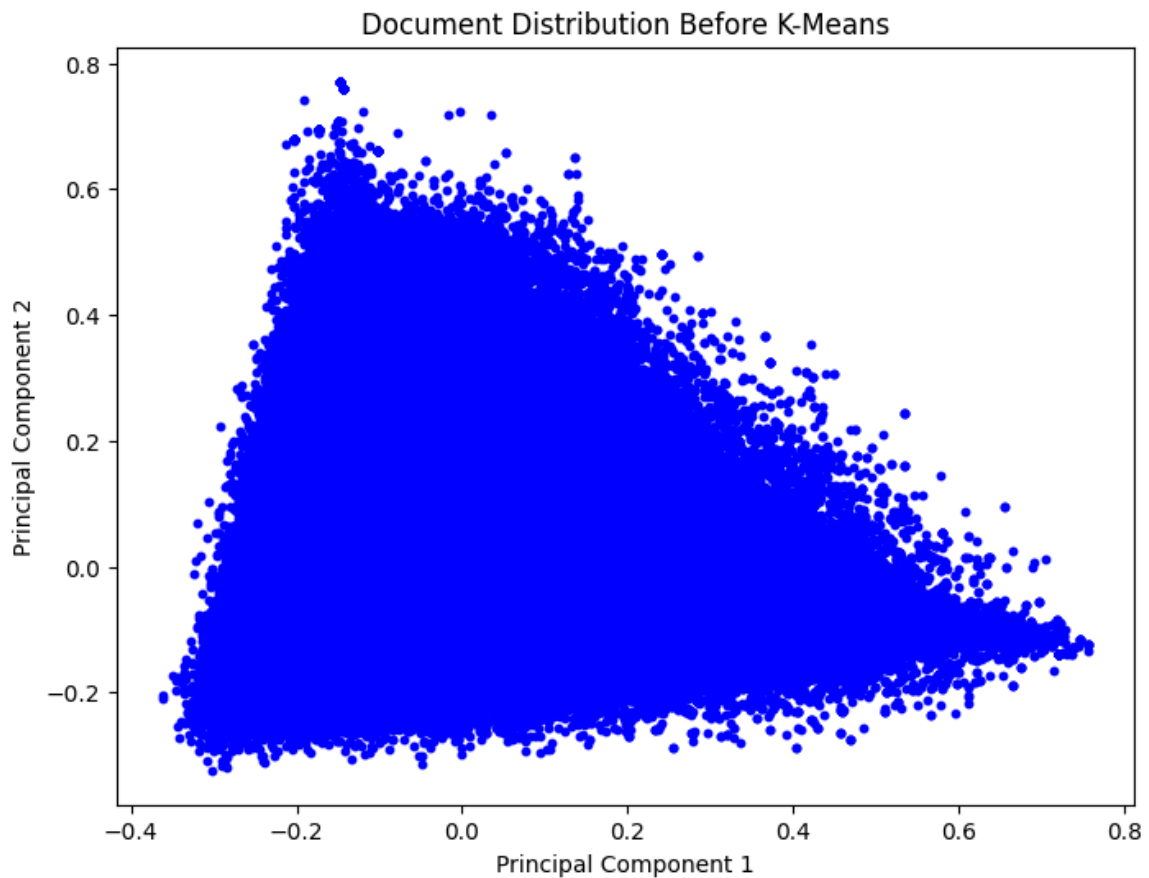
✓ Before use K-mean

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(max_features=1000)
X = vectorizer.fit_transform(df['sum'])
```

```
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
# PCA 2D
pca = PCA(n_components=2, random_state=42)
X_reduced = pca.fit_transform(X.toarray())

# Scatter plot ก่อน K-Means
plt.figure(figsize=(8,6))
plt.scatter(X_reduced[:,0], X_reduced[:,1], s=10, color='blue')
plt.title("Document Distribution Before K-Means")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()
```



✓ After use K-mean (K=5)

```
from sklearn.cluster import KMeans

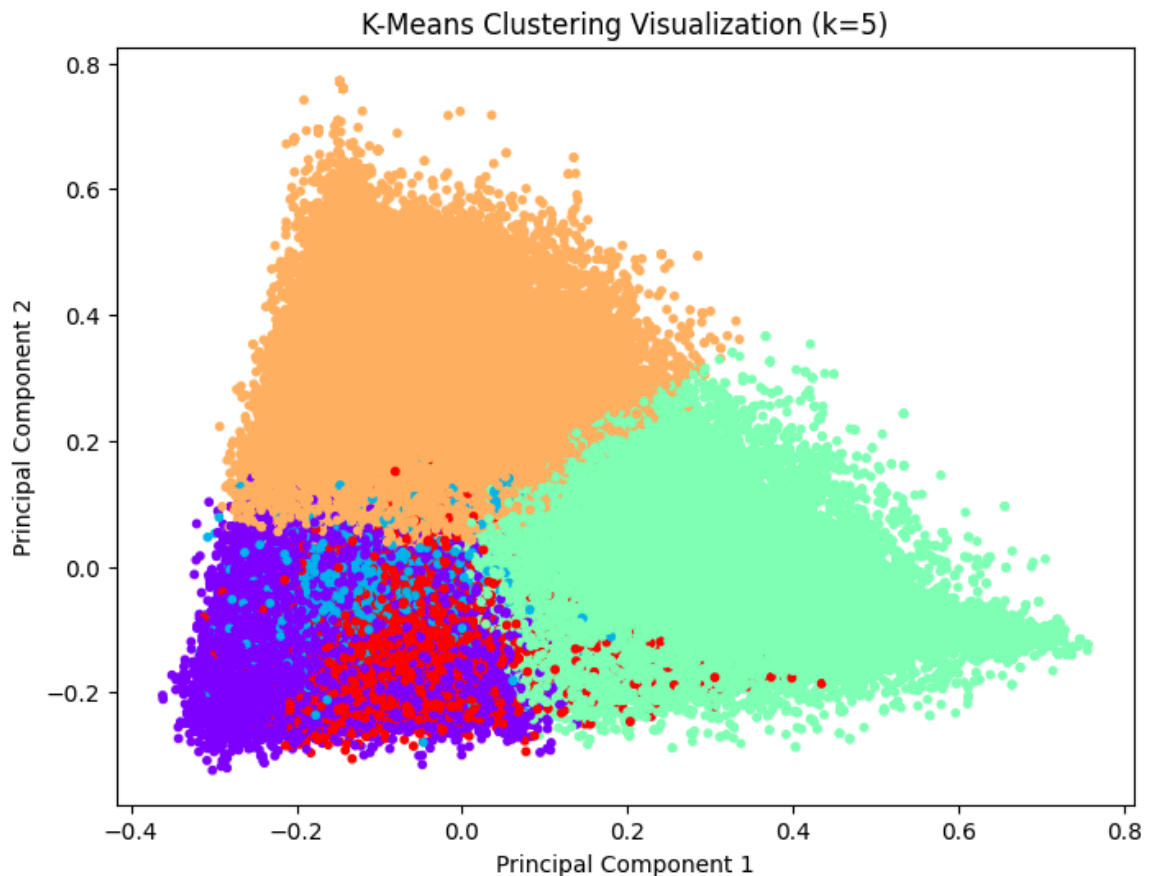
k = 5
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
kmeans.fit(X)

df['cluster'] = kmeans.labels_
```

```
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# ลดมิติ TF-IDF เหลือ 2 มิติ
pca = PCA(n_components=2, random_state=42)
X_reduced = pca.fit_transform(X.toarray())

# Scatter plot ตาม cluster
plt.figure(figsize=(8,6))
plt.scatter(X_reduced[:,0], X_reduced[:,1], c=df['cluster'], cmap='rainbow', s=10)
plt.title(f"K-Means Clustering Visualization (k={k})")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()
```



```
terms = vectorizer.get_feature_names_out()
order_centroids = kmeans.cluster_centers_.argsort()[:, ::-1]

for i in range(k):
    top_terms = [terms[ind] for ind in order_centroids[i, :10]] # top 10 terms
    print(f"\nCluster {i}: {' '.join(top_terms)}")
```

Cluster 0: loan, payment, mortgage, pay, call, make, would, student, tell, get

Cluster 1: account, bank, check, money, saving, deposit, transfer, fee, fund, call

Cluster 2: report, credit, consumer, account, personal, repair, service, information, remove, inquiry

Cluster 3: debt, collection, call, credit, company, report, account, owe, receive, pay

Cluster 4: card, credit, charge, account, payment, call, pay, prepay, bank, balance

```
# แสดงตัวอย่างข้อความในแต่ละ cluster
for i in range(k):
    print(f"\nCluster {i} examples:")
    print(df['sum'][df['cluster']==i].head(5).to_list())
```

Cluster 0 examples:

['mortgage already try contact company following complaint time use different avenue recent email send

Cluster 1 examples:

['money transfer virtual currency money service sell access event digitally screenshots detail transac

Cluster 2 examples:

['credit report credit repair service personal consumer report would like request suppression followin

Cluster 3 examples:

['debt collection transworld system inc try collect debt mine owe inaccurate', 'debt collection past w

Cluster 4 examples:

['credit report credit repair service personal consumer report month make daughter authorize user acco

✓ After use K-mean (K= 3, 4, 7)

k = 3

```
from sklearn.cluster import KMeans
```

```
k = 3
```

```
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
```

```
kmeans.fit(X)
```

```
df['cluster'] = kmeans.labels_
```

```
# ลดมิติ TF-IDF เหลือ 2 มิติ
```

```
pca = PCA(n_components=2, random_state=42)
```

```
X_reduced = pca.fit_transform(X.toarray())
```

```
# Scatter plot ตาม cluster
```

```
plt.figure(figsize=(8,6))
```

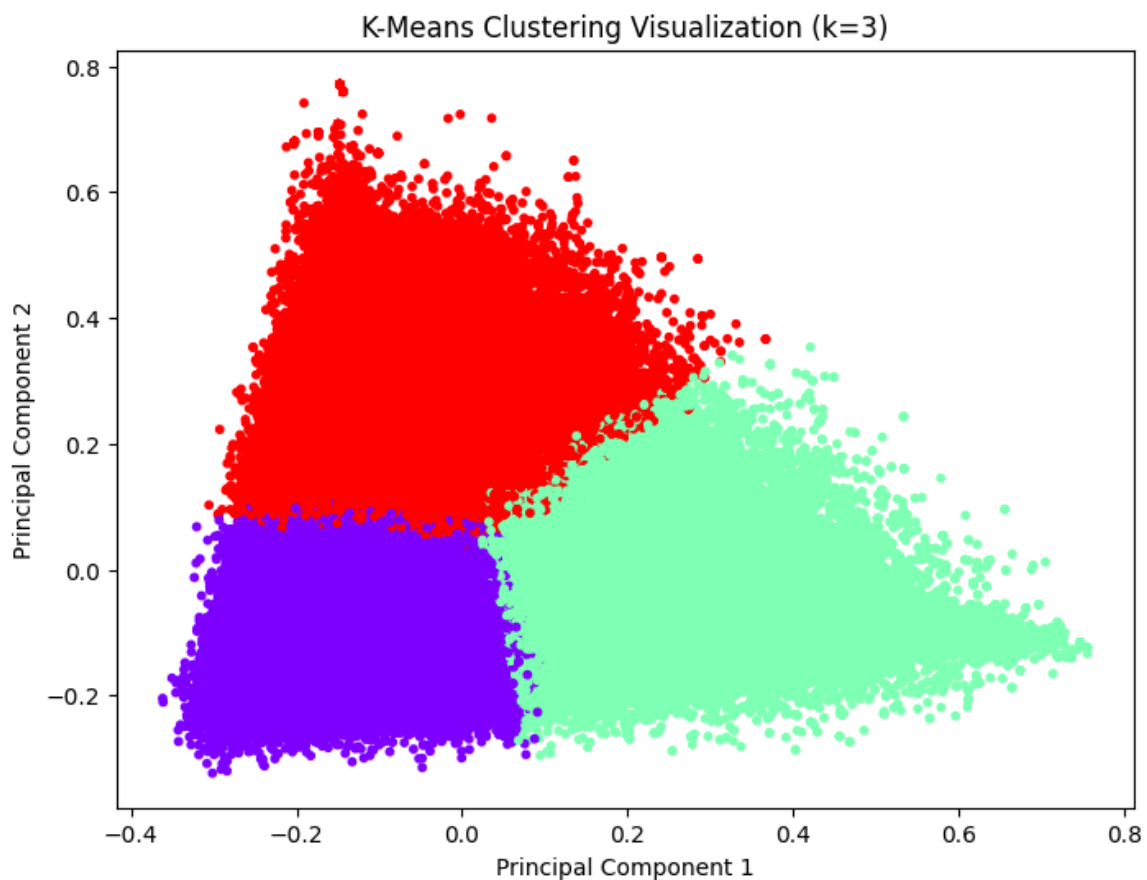
```
plt.scatter(X_reduced[:,0], X_reduced[:,1], c=df['cluster'], cmap='rainbow', s=10)
```

```
plt.title(f"K-Means Clustering Visualization (k={k})")
```

```
plt.xlabel("Principal Component 1")
```

```
plt.ylabel("Principal Component 2")
```

```
plt.show()
```



```
terms = vectorizer.get_feature_names_out()
```

```
order_centroids = kmeans.cluster_centers_.argsort()[:, :-1]
```

```
for i in range(k):
```

```
top_terms = [terms[ind] for ind in order_centroids[i, :10]] # top 10 terms
print(f"\nCluster {i}: {' '.join(top_terms)}")
```

Cluster 0: loan, payment, card, account, bank, mortgage, pay, call, credit, tell

Cluster 1: report, credit, consumer, account, personal, repair, service, information, remove, inquiry

Cluster 2: debt, collection, call, credit, company, report, account, owe, receive, letter

```
# แสดงตัวอย่างข้อความในแต่ละ cluster
for i in range(k):
    print(f"\nCluster {i} examples:")
    print(df['sum'][df['cluster']==i].head(5).to_list())
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Cluster 1 examples:

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Cluster 2 examples:

['debt collection transworld system inc try collect debt mine owe inaccurate', 'debt collection past w

```
from sklearn.cluster import KMeans
```

```
k = 4
```

```
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
```

```
kmeans.fit(X)
```

```
df['cluster'] = kmeans.labels_
```

k = 4

```
# ลดมิติ TF-IDF เหลือ 2 มิติ
```

```
pca = PCA(n_components=2, random_state=42)
```

```
X_reduced = pca.fit_transform(X.toarray())
```

```
# Scatter plot ตาม cluster
```

```
plt.figure(figsize=(8,6))
```

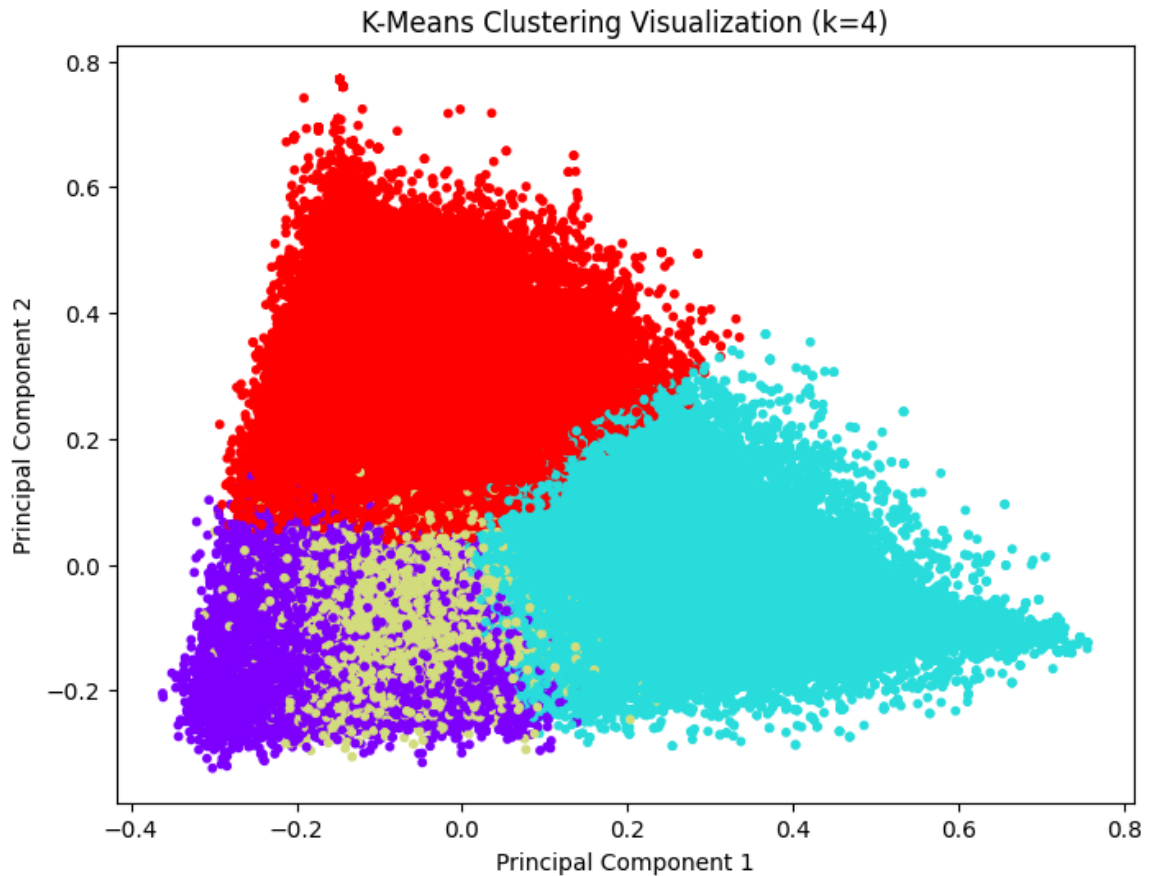
```
plt.scatter(X_reduced[:,0], X_reduced[:,1], c=df['cluster'], cmap='rainbow', s=10)
```

```
plt.title(f"K-Means Clustering Visualization (k={k})")
```

```
plt.xlabel("Principal Component 1")
```

```
plt.ylabel("Principal Component 2")
```

```
plt.show()
```



```
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    top_terms = [terms[ind] for ind in order_centroids[i, :10]] # top 10 terms
    print(f"\nCluster {i}: {' '.join(top_terms)}")
```

Cluster 0: loan, payment, mortgage, pay, call, make, would, student, tell, get

Cluster 1: report, credit, consumer, account, personal, repair, service, information, remove, inquiry

Cluster 2: card, account, bank, credit, charge, check, money, call, fee, tell

Cluster 3: debt, collection, call, credit, company, report, account, owe, receive, pay

```
# แสดงตัวอย่างข้อความในแต่ละ cluster
for i in range(k):
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Cluster 3 examples:

['debt collection transworld system inc try collect debt mine owe inaccurate', 'debt collection past w

k = 7

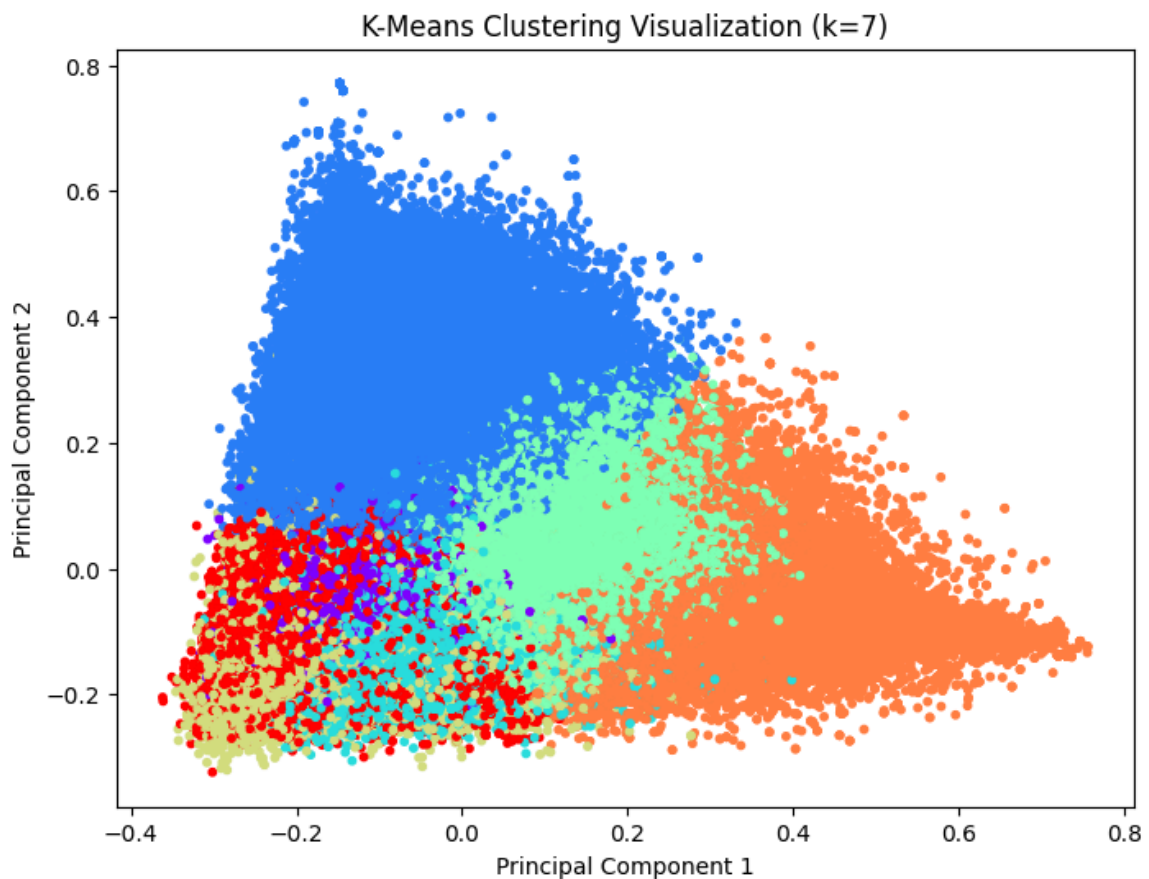

```
from sklearn.cluster import KMeans

k = 7
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
kmeans.fit(X)

df['cluster'] = kmeans.labels_
```

```
# ลดมิติ TF-IDF เหลือ 2 มิติ
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```



```
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    print(f"\nCluster {i}: {' '.join(top_terms)}")
```

Cluster 0: account, bank, check, money, saving, deposit, transfer, fee, fund, call

Cluster 1: debt, collection, call, company, credit, report, owe, account, pay, receive

Cluster 2: card, credit, charge, account, payment, call, pay, prepay, bank, balance

Cluster 3: credit, report, account, information, reporting, dispute, equifax, consumer, remove, person

Cluster 4: loan, student, payment, pay, navient, interest, make, call, would, month

Cluster 5: report, credit, repair, personal, consumer, service, inquiry, account, remove, information

Cluster 6: mortgage, payment, loan, pay, call, home, make, would, tell, get